ELTE
FACULTY OF INFORMATICS

Patterns

Summation

Let an interval of integers [b..e] and a function f: [b..e]→H be given. Let us suppose that the addition operation is defined on the elements of H.

Determine the sum of the values of the f function on the interval [b..e], i.e. the value of the expression $\sum_{i=b}^{e} f(i)$ (if b>e the value will be 0)

Specification:

```
In: b∈Z, e∈Z
Out: s∈H
Pre: -
Post: s=SUM(i=b..e, f(i))
```

Algorithm:

```
s:=0
i=b..e
    s:=s+f(i)
```

Counting

Let an interval of integers [b..e] and a conditon A: [b..e]→L be given.

Determine the number of times A attains the "true" value on the interval [b..e] (in the case of an empty interval, if b>e the value returned will be 0)

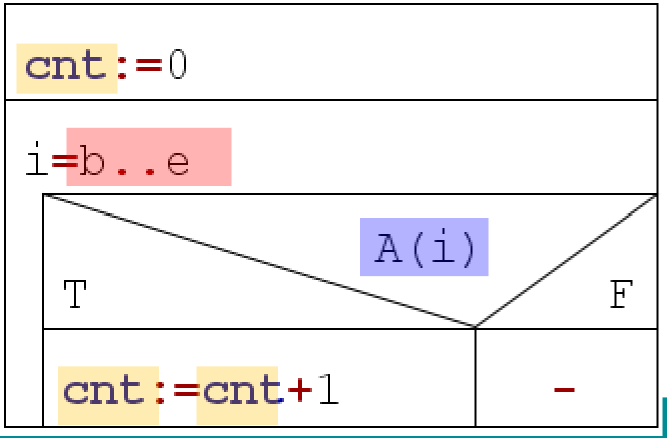
Specification:

```
In: b∈Z, e∈Z
Out: cnt∈N
Pre: -
Post: cnt=SUM(i=b..e, 1,A(i))
```

shorting:

```
Post: cnt=COUNT(i=b..e,A(i))
```

Algorithm:



Maximum selection

Let a non-empty interval of integers [b..e] and a function f: [b..e]→H be given. Let us suppose that a total order is defined on the elements of H.

Determine the point (index) at which attains its maximum value on the interval [b..e] and calculate this **value** too. (if there is more than one maximum, any of them can be selected)

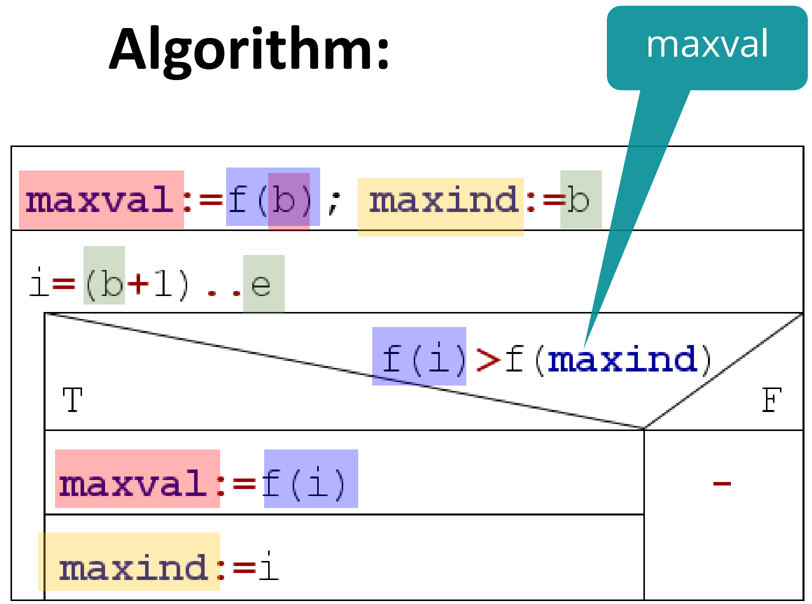
Specification:

```
In: b∈Z, e∈Z
Out: maxind∈Z, maxval∈H
Pre: b<=e
Post: maxind∈[b..e] and
      ∀i∈[b..e]:(f(maxind)>=f(i)) and
      maxval=f(maxind)
```

shorting:

```
Post:
(maxind,maxval)=MAX(i=b..e,f(i))
```

Algorithm:



Search

Let an interval of integers [b..e] and a condition A: [b..e]→L be given.

Determine the ***first*** point (index) starting from left at which A attains the "true" value on the interval [b..e] if it exists. (The empty interval is an equivalent of A evaluating to "false" on the whole interval.)

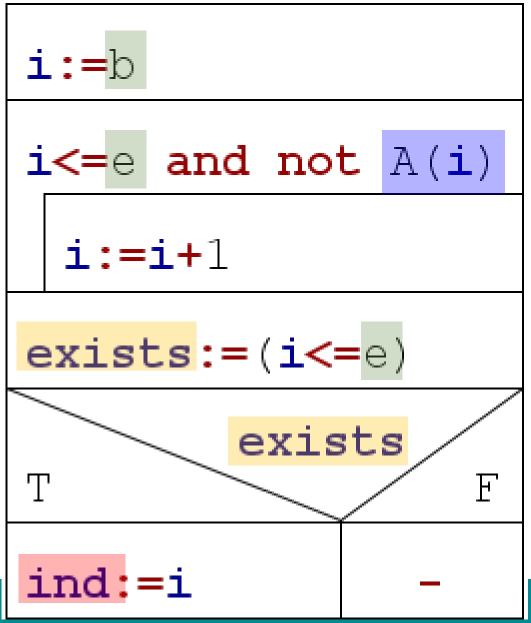
Specification:

```
In: b∈Z, e∈Z
Out: exists∈L, ind∈Z
Pre: -
Post: exists=∃i∈[b..e]:(A(i)) and
       exists->(ind∈[b..e] and A(ind) and
          ∀i∈[b..ind-1]:(not A(i)))
```

shorting:

```
Post: (exists,ind)=SEARCH(i=b..e,A(i))
```

Algorithm:



Decision

Let an interval of integers [b..e] and a condition A: [b..e]→L be given.

Determine whether it exists a point at which A attains the "true" value on the interval [b..e]. (The empty interval is an equivalent of A evaluating to "false" on the whole interval.)

Specification:

In: b∈Z, e∈Z

Out: exists∈L, ind∈Z

Pre: -

Post: exists=∃i∈[b..e]:(A(i))

shorting:

Post: exists=**EXISTS**(i=b..e,A(i))

```
i:=b
i<=e and not A(i)
    i:=i+1
exists:=(i<=e)
```

Selection

Let an interval of integers [b..e] and a condition A: [b..e]→L be given.

Determine the leftmost point starting with b at which A attains the "true" value on the interval [b..e] if we know that such point definitely exists.

Specification:

```
In: b∈Z, e∈Z
Out: ind∈Z
Pre: ∃i∈[b..e]:(A(i))
Post: ind>=b and A(ind) and
       ∀i∈[b..ind-1]:(not A(i)))
```

shorting:

```
Post: ind=SELECT(i>=b,A(i))
```

```
ind:=b
not A(ind)
    ind:=ind+1
```

Copy

Let an interval of integers [b..e] and a function f: [b..e]$\rightarrow$H be given.

Assign to every element (point) on the interval [b..e] the value imaged by the f function

Specification:

In: b∈Z, e∈Z

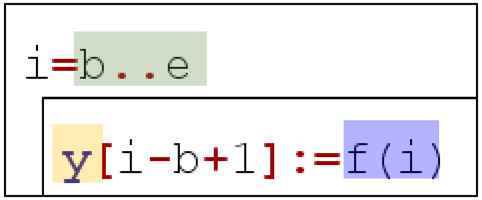
Out: y∈H[1..e-b+1]

Pre: -

Post: ∀i∈[b..e]:(y[i-b+1]=f(i))

shorting:

Post: y=**COPY**(i=b..e,f(i))



Multiple item selection

i		A(i)			f(i)
b	→	false			
b+1	→	true	→	1	f(b+1)
b+2	→	true	→	2	f(b+2)
e	→	false			

Let an interval of integers [b..e], a condition A: [b..e]→L and a function f: [b..e]→H be given.

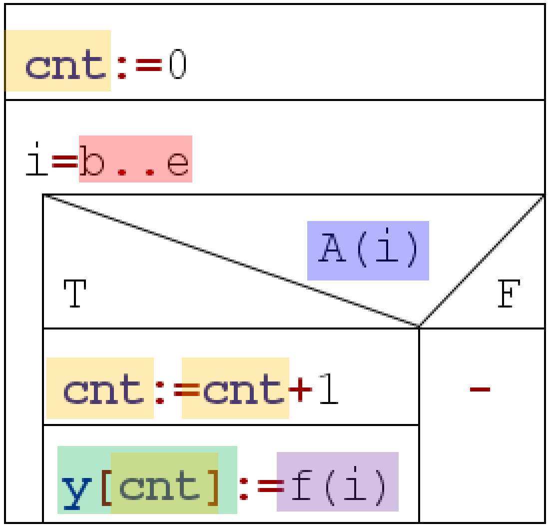
Determine the f values of the point(s) at which A attains the "true" value on the interval [b..e]

Specification:

```
In: b∈Z, e∈Z
Out: cnt∈N, y∈H[1..cnt]
Prec: -
Post: cnt=COUNT(i=b..e,A(i)) and
      ∀i∈[1..cnt]:(∃j∈[b..e]:A(j) and
       y[i]=f(j)) and
       y⊆(f(b),f(b+1),...,f(e))
```

shorting:

```
Postc: (cnt,y)=MIS(i=b..e,A(i),f(i))
```



Minimum selection

Let a non-empty interval of integers [b..e] and a function f: [b..e]→H be given. Let us suppose that a total order is defined on the elements of H.

Determine the point (index) at which attains its minimum value on the interval [b..e] and calculate this **value** too. (if there is more than one maximum, any of them can be selected)

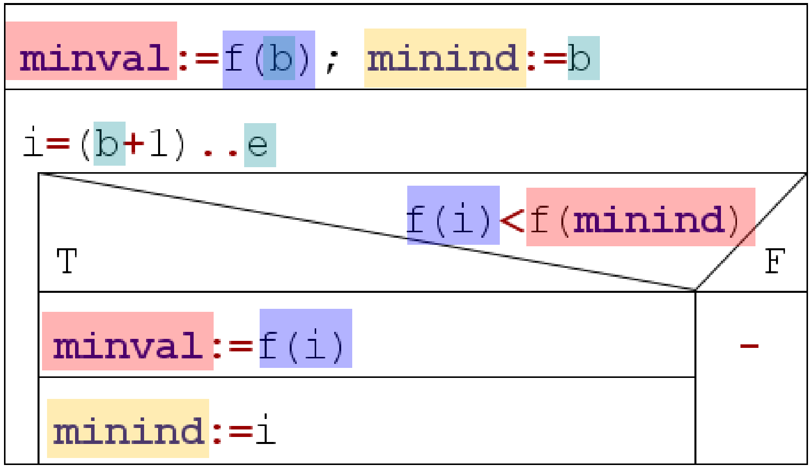
Specification:

```
In: b∈Z, e∈Z
Out: minind∈Z, minval∈H
Pre: b<=e
Post: minind∈[b..e] and
      ∀i∈[b..e]:(f(minind)<=f(i)) and
      minval=f(minind)
```

shorting:

```
Post:
(minind,minval)=MIN(i=b..e,f(i))
```

Algorithm:



Decision all

Let an interval of integers [b..e] and a condition A: [b..e]→L be given.

Determine whether A attains the "true" value on the interval [b..e] at all points.

Specification:

```
In: b∈Z, e∈Z
Out: all∈L
Pre: -
Post: all= ∀i∈[b..e]: A(i)
```

shorting:

```
Post: all=FORALL(i=b..e,A(i))
```

Algorithm:

```
i:=b
i<=e and A(i)
    i:=i+1
all:=(i>e)
```